INTERNSHIP OFFER (4 to 6 months) • 2027

# Investigating control strategies in multistable dynamical systems: the example of bowed string musical instruments

This internship is funded by the French National Research Agency (ANR) as part of the BEBOP project (Beyond Classical Bifurcation Analysis: New Perspectives for Mechanical Engineering).
**A fully-funded PhD building on this topic is available after the internship.**

## Context

**Nonlinear dynamical systems** such as pulsing lasers, musical instruments or climate models, are known to exhibit **complex behaviours**. This includes the coexistence of multiple stable regimes for identical parameters (a situation referred to as multistability), a possible sensitivity to initial conditions or sudden transitions between regimes under the small change of a parameter. **Understanding and predicting** the dynamics of such nonlinear systems remains a major scientific challenge.

**Numerical bifurcation analysis** has attracted considerable attention over the last decades, and lead to substantial progresses by giving access to **bifurcation diagrams** which represent ideally all the possible stable and unstable solutions of a system, with respect to a parameter of interest. Nevertheless, bifurcation analysis also has important limitations. In particular, it unveils the **multistability regions** of the system (the range of parameters for which several stable solutions coexist) but says nothing on which regime (solution) is reached in practice.

Self-sustained musical instruments, such as wind instruments and bowed string instruments, are an excellent testbed to explore general phenomena in nonlinear dynamics. Indeed, sound production in these instruments relies on a wide diversity of nonlinear mechanisms in connexion with various scientific fields (contact mechanics, aeroacoustics, biomechanics…). Such instruments have also been shown to produce a wealth of sound regimes whose properties and existence depend - often sensitively - on **design and playing parameters.**

From a musical point of view, **multistability regions** are particularly interesting as they allow the musician to access a wide range of sound regimes, while also posing challenges in controlling them. **In practice, how to access a particular sound regime amongst the different possible ones?** From a dynamical systems perspective, this raises the question of **basins of attraction**, that is to say the question of initial conditions that make the system converge towards a particular solution.

Recent work at LAUM and LMA [1,2] focused on developing numerical tools to compute **borders between basins of attraction**, with application to physical models of musical instruments. In parallel, preliminary work [3,4] have demonstrated that the **temporal evolution (dynamics) of control parameters** can play a crucial role in determining which regime is observed, and even trigger unexpected transitions between regimes. Such rapid variations of parameters are observed for example during the so-called **attack transients,** when a musician starts to play a note and varies rapidly the contact force between the bow and the string and/or the bow speed. These observations relate to the phenomenon of **rate-induced tipping,** which has attracted considerable attention recently, particularly in the context of climate modelling [5].

This internship is part of the project BEBOP, funded by the French National Research Agency (ANR). This project aims at exploring the dynamics of nonlinear systems beyond the classical bifurcation analysis, in order to better understand and predict the complex dynamics of nonlinear systems in general, with a focus on self-sustained musical instruments.

## Objectives

The proposed work aims at studying, in systems subject to friction-induced vibrations (such as bowed-string musical instruments), control strategies allowing to access a particular solution amongst the different possible ones (for example, production of sound or not). A particular focus will be put on investigating the influence of **small perturbations on both initial conditions and control parameters** - such as the bow speed - on **transitions between different regimes** (corresponding for example to sound production or silence).

## Methodology

The proposed approach combines **numerical simulations and experiments.** It will focus on a simple academic system, which is nevertheless representative of the production of a given note by a bowed-string instrument such as the violin.

A **first work axis** will aim at characterise the *resilience* of the model, that is to say its **ability to resist to perturbations** without changing regime. Numerically, both the influence of perturbations on state variables (in relation with basins of attraction) and on control parameters (bow speed and bow normal force) with be investigated. By doing so, we will aim at **understanding the connexion between basins of attraction and the control strategies of musicians**: can a performer use perturbations on parameters to control indirectly initial conditions and therefore the observed regime ?

A **second work axis** will aim at exploring these questions experimentally. A controlled testbench already available at LAUM, which reproduces the stick-slip phenomenon at the origin of sound production in bowed-string instruments, will be used to study systematically the effect of perturbations on both control parameters and variables.

## Profile of the candidate

Master (final year) student in mechanical engineering, acoustics or nonlinear physics. A background in nonlinear dynamics, in scientific programming, and/or in experiments in mechanics/acoustics will be appreciated. An interest for musical instruments will be a plus.

## Environnement and supervision

This internship will take place at the **Laboratoire d'Acoustique de l'Université du Mans (LAUM)**, a research lab of CNRS and Le Mans Université. It will be supervised by Soizic Terrien, research scientist at CNRS, and Frédéric Ablitzer, lecturer at Le Mans Université. This internship is part of a collaborative project with Baptiste Bergeot, lecturer at INSA Centre Val de Loire/Lamé (Blois, France) and Christophe Vergez, CNRS researcher at the Laboratoire de Mécanique et d'Acoustique (Marseille, France), with whom regular interactions will be organised. Remuneration follows CNRS rules (about 650 euros/month).
**A fully-funded PhD position on related topics is possible after the internship.**

## How to apply

CV, cover letter, academic transcript and contact of academic referees should be sent to Soizic Terrien (soizic.terrien@univ-lemans.fr) no later than 1st of December 2026. Starting date can be any time from 1st of February 2027 and total duration can be between 4 and 6 months.

## References


[1] Passa, T., Terrien, S., Maugeais, S., & Gazengel, B. (2025). Basins of attraction of periodic solutions in a multistable model of self-oscillating musical instrument. *Journal of Sound and Vibration*, 119467.

[2] Pégeot, M., Colinot, T., Doc, J. B., Fréour, V., & Vergez, C. (2024). Playability of self-sustained musical instrument models: statistical approaches. *Acta Acustica*, *8*, 74.

[3] Bergeot, B., Terrien, S., & Vergez, C. (2024). Predicting transient dynamics in a model of reed musical instrument with slowly time-varying control parameter. *Chaos*, *34*(7).

[4] Terrien, S., Bergeot, B., Vergez, C., & Missoum, S. (2025). Basins of attraction in a dynamical system with a time-varying control parameter: The case of attack transients in a simple model of reed musical instrument. *Journal of Sound and Vibration*, *618*, 119241.

[5] Ritchie, P. D., Alkhayuon, H., Cox, P. M., & Wieczorek, S. (2023). Rate-induced tipping in natural and human systems. *Earth System Dynamics*, *14*(3), 669-683.